# LAPORAN PRAKTIKUM

# POSTTEST (2)

# ALGORITMA PEMROGRAMAN LANJUT

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**PROGRAM STUDI INFORMATIKA**

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**SAMARINDA**

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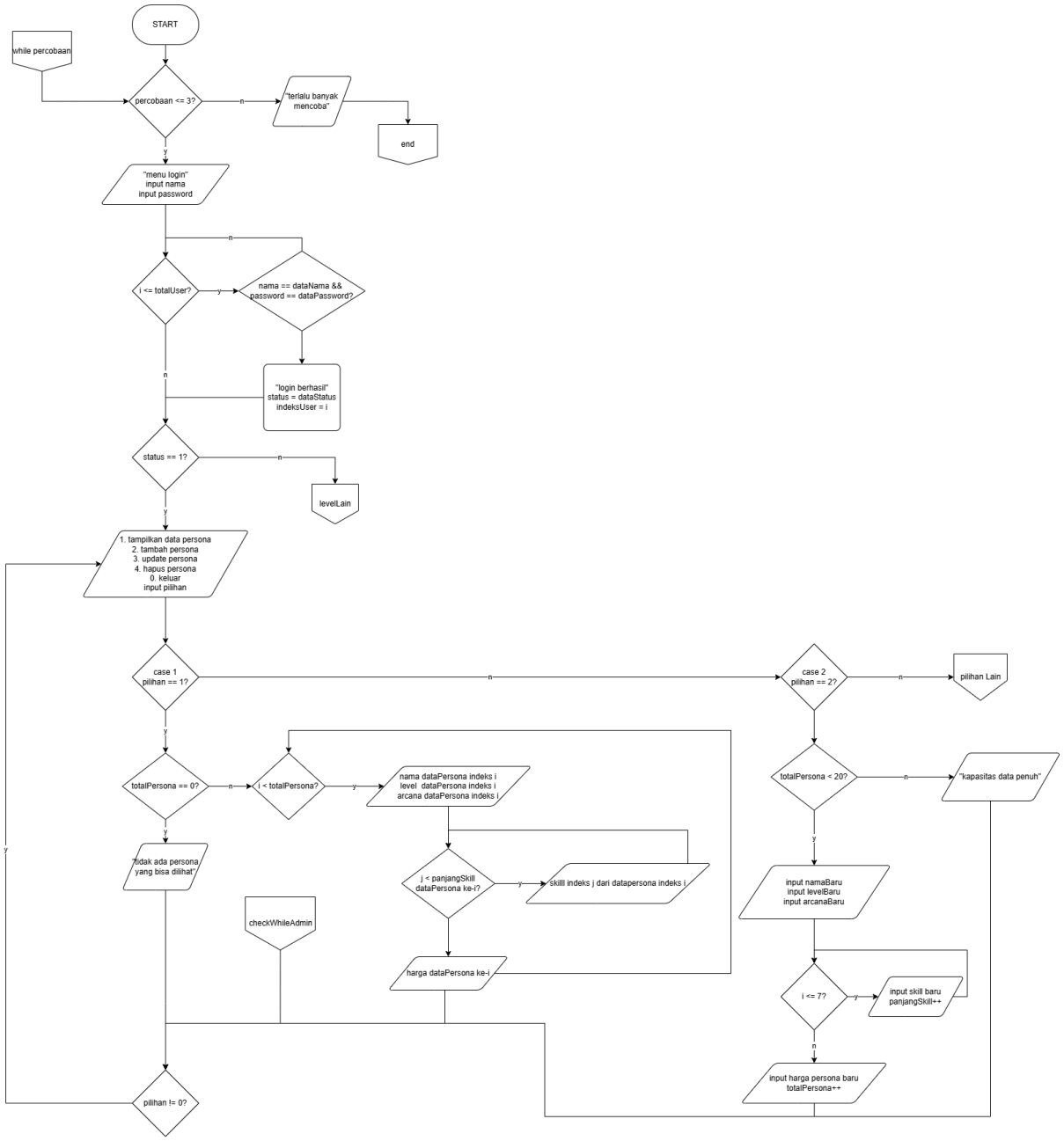
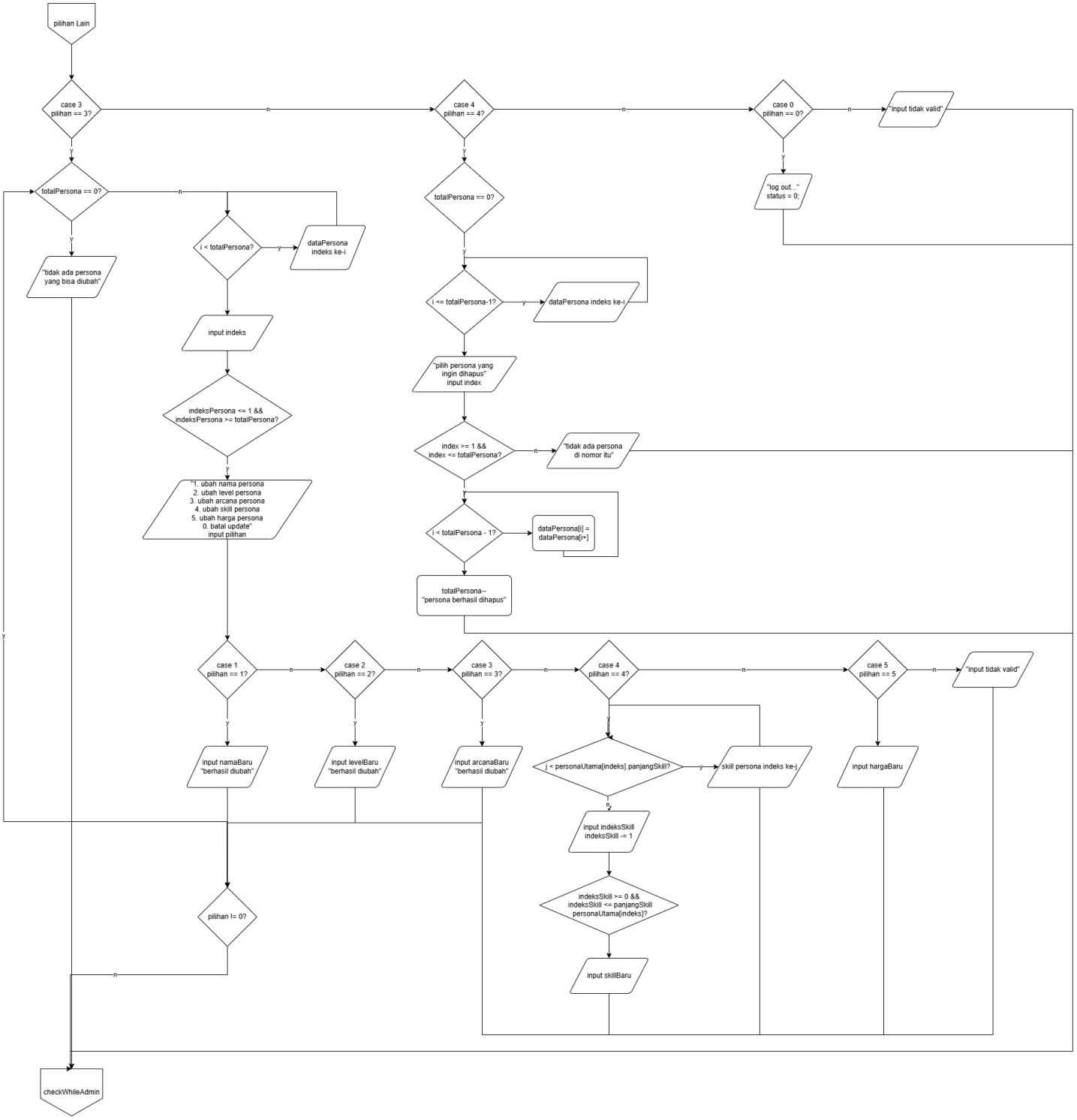
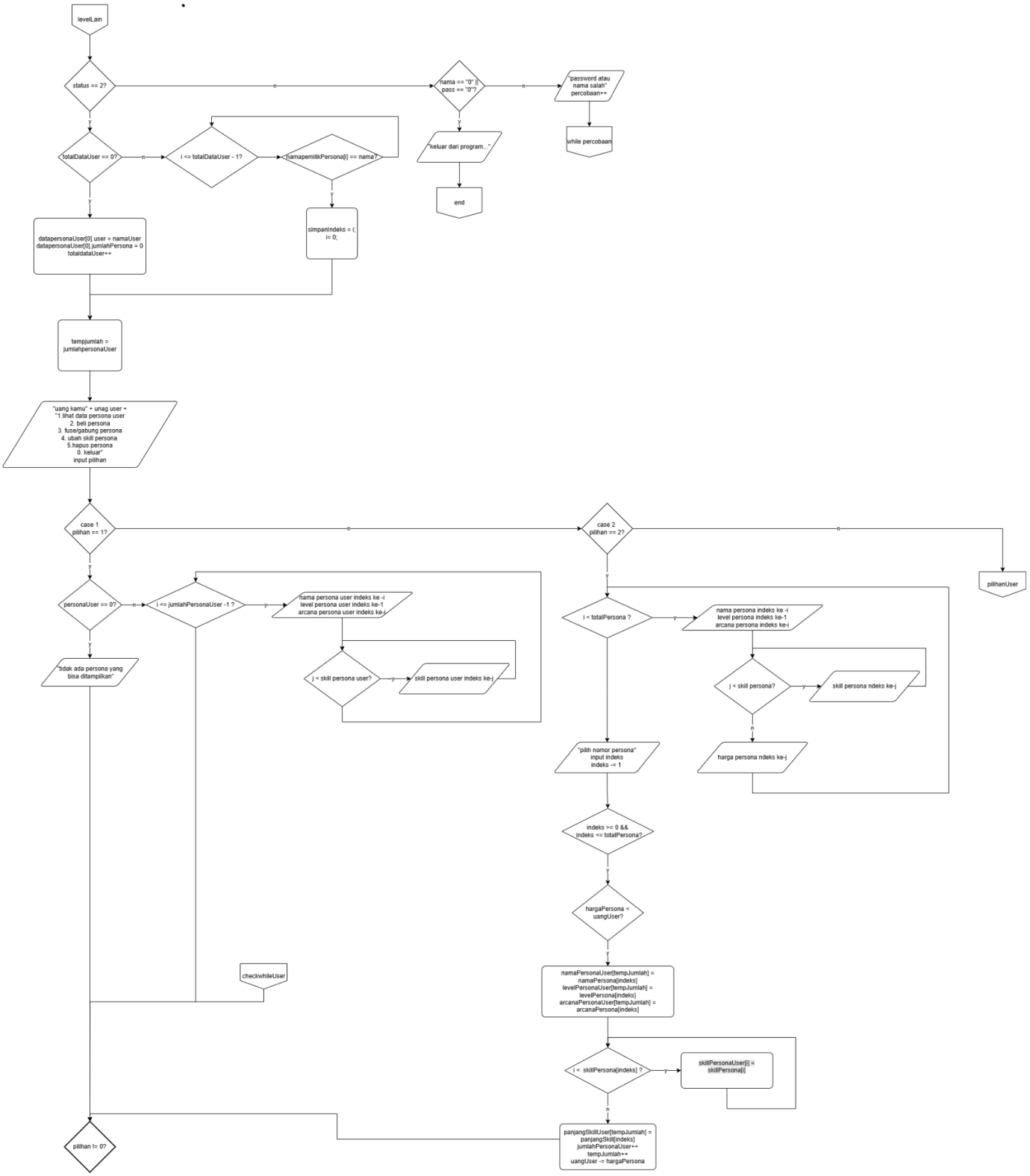
# 1. Flowchart



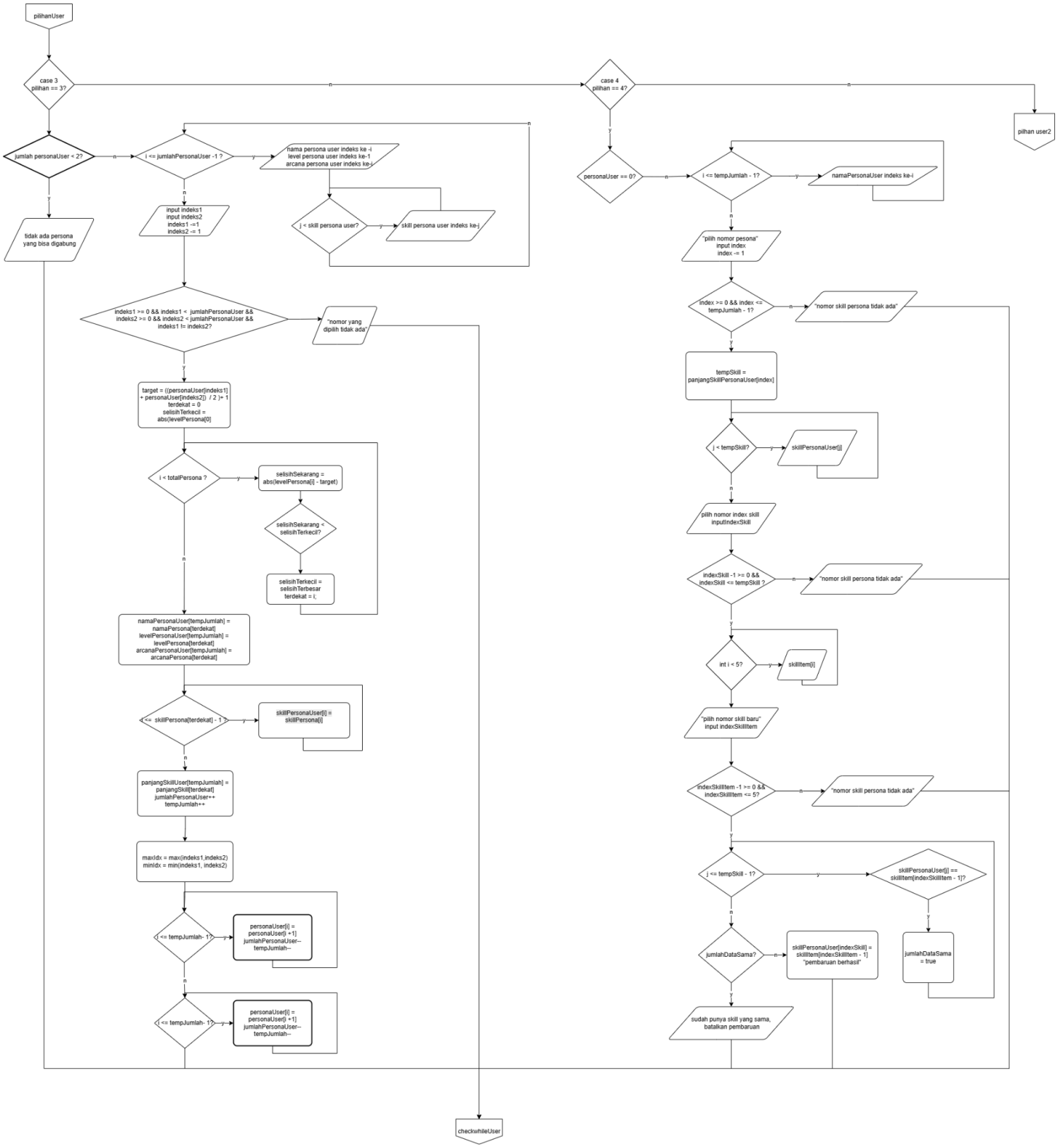

gambar 1.1 flowchart login dan opsi 1-2 admin



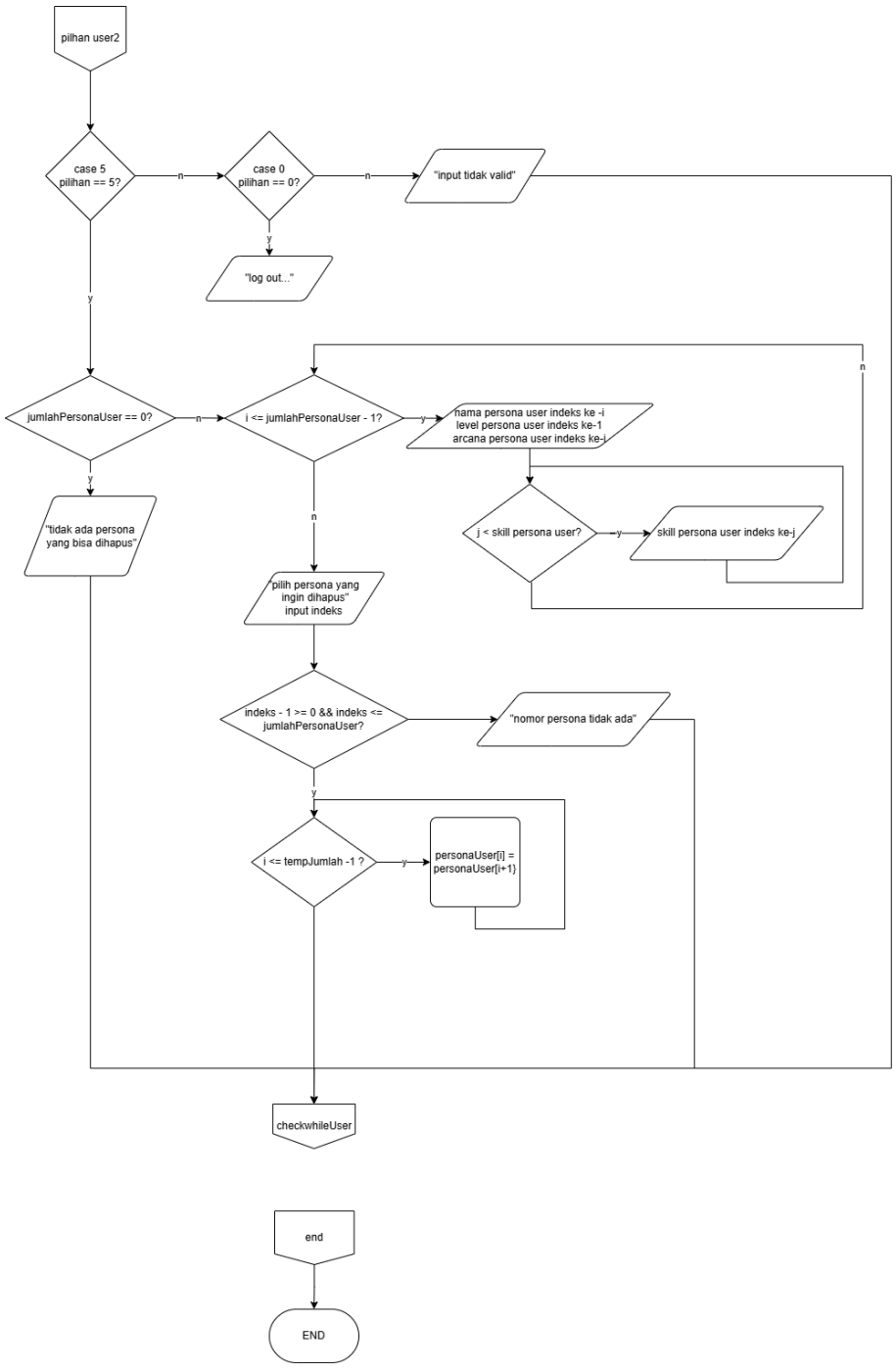

gambar 1.2 flowchart opsi 3 sampai log out admin




gambar 1.3 flowchart opsi user 1 -2




gambar 1.4 flowchart opsi user 3-4




gambar 1.5 flowchart opsi user ke-5

## 2. Deskripsi Singkat Program

pada posttest ini saya mencoba membuat velvet room pada game Persona 3 Reload. disini, terdapat 2 level user, dimana user biasa bisa melihat, membeli, menggabungkan, edit skill, dan hapus persona, sedangkan untuk admin, admin bisa membuat persona baru, melihat, mengedit, dan menghapus persona yang dipakai untuk user nanti.

## 3. Source Code

### A. fitur user

code ini terdapat semua fitur user, mulai dari lihat persona user hingga menghapus nya

**Source Code:**

```
else if(status == 2){
            percobaan = 0;
            string simpanNama = inputNama;
            int simpanIndeks = 0;
            if (totalDataUser == 0){
                dataPersonaUser[0].user = simpanNama;
                dataPersonaUser[0].jumlahPersona = 0;
                simpanIndeks = 0; //mastiin
                totalDataUser++;

            }else {
                int i = 0;
                while (i <= totalDataUser - 1)
                {
                    if(dataPersonaUser[i].user == simpanNama){
                        simpanIndeks = i;
                        i = 0;
                        break;
                    }
                    i++;
                }
            }
            int tempJumlah = dataPersonaUser[simpanIndeks].jumlahPersona;
            do {
                cout << "\n = menu velvet room user =" << endl;
                cout << "uang kamu : " << user[indeksUser].uang << " yen"<< endl;
                cout << "1. lihat data persona user" << endl;
                cout << "2. beli persona" << endl;
                cout << "3. fuse/gabung persona" << endl;
```

```
                cout << "4. ubah skill persona" << endl;
                cout << "5. hapus persona" << endl;
                cout << "0. keluar" << endl;
                cout << "Pilihan: ";
                cin >> pilihan;

                switch (pilihan)
                {
                    case 1:
                    if (dataPersonaUser[simpanIndeks].jumlahPersona == 0)
                    {
                        cout << "tidak ada persona yang dilihat" << endl;
                    }else {
                        for(int i = 0; i <=
dataPersonaUser[simpanIndeks].jumlahPersona - 1; i++){

                            cout << "\n ======" <<
dataPersonaUser[simpanIndeks].listPersona[i].nama << "======" << endl;
                            cout << "level    : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].level << endl;
                            cout << "arcana   : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].arcana << endl;

                            cout << "skill    : ";
                            for (int j = 0; j <
dataPersonaUser[simpanIndeks].listPersona[i].panjangSkill ; j++){
                                cout <<
dataPersonaUser[simpanIndeks].listPersona[i].skill[j] << ", ";
                            }
                            cout << endl;
                        }
                    }

                    break;
                    case 2:
                    for(int i = 0; i < totalPersona ; i++){
                        cout << "\n ======" << i + 1 << ". " << dataPersona[i].nama
<< "======" << endl;
                        cout << "level    : "  << dataPersona[i].level << endl;
                        cout << "arcana   : "  << dataPersona[i].arcana << endl;
                        cout << "skill    : ";
                        for (int j = 0; j < dataPersona[i].panjangSkill ; j++){
                            cout << dataPersona[i].skill[j] << ", ";
                        }
                        cout << endl;
                        cout << "harga   : "  << dataPersona[i].harga << endl;
                    }

                    cout << "pilih nomor persona yang ingin dibeli :"; cin >> index;
                    index -= 1;
                    if(index >= 0 && index <= totalPersona){
                        if(dataPersona[index].harga < user[indeksUser].uang){
```

```
dataPersonaUser[simpanIndeks].listPersona[tempJumlah].nama =
dataPersona[index].nama;

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].level =
dataPersona[index].level;

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].arcana =
dataPersona[index].arcana;
                            for(int i = 0; i <= dataPersona[index].panjangSkill - 1;
i++){

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].skill[i] =
dataPersona[index].skill[i];
                            }

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].panjangSkill =
dataPersona[index].panjangSkill;
                            dataPersonaUser[simpanIndeks].jumlahPersona++;
                            tempJumlah++;

                            user[indeksUser].uang -= dataPersona[index].harga;
                            cout << "persona berhasil dibeli" << endl;

                        }
                    }
                    break;
                    case 3:

                    if (dataPersonaUser[simpanIndeks].jumlahPersona < 2)
                    {
                        cout << "tidak ada persona yang bisa digabung" << endl;
                    }else {
                        for(int i = 0; i <=
dataPersonaUser[simpanIndeks].jumlahPersona - 1; i++){

                            cout << "\n ======" << i + 1 << ". "<<
dataPersonaUser[simpanIndeks].listPersona[i].nama << "======" << endl;
                            cout << "level    : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].level << endl;
                            cout << "arcana   : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].arcana << endl;

                            cout << "skill    : ";
                            for (int j = 0; j
<dataPersonaUser[simpanIndeks].listPersona[i].panjangSkill - 1; j++){
                                cout <<
dataPersonaUser[simpanIndeks].listPersona[i].skill[j] << ", ";
                            }
                    cout << endl;
                }
            }

            int persona1, persona2;
```

```
            cout << "masukkan persona pertama yang ingin digabung "; cin >>
persona1;
            cout << "masukkan persona kedua yang ingin digabung "; cin >> persona2;
            persona1 -= 1;
            persona2 -= 1;

            if((persona1 >= 0 && persona1 <
dataPersonaUser[simpanIndeks].jumlahPersona) && (persona2 >= 0 && persona2 <
dataPersonaUser[simpanIndeks].jumlahPersona) && persona1 != persona2 ){
                int target =
((dataPersonaUser[simpanIndeks].listPersona[persona1].level +
dataPersonaUser[simpanIndeks].listPersona[persona2].level) / 2 ) + 1;

                int terdekat = 0;
                double selisihTerkecil = abs(dataPersona[0].level);

                for (int i = 0; i < totalPersona ; i ++){
                    double selisihSekarang = abs(dataPersona[i].level - target);
                    if (selisihSekarang < selisihTerkecil) {
                        selisihTerkecil = selisihSekarang;
                        terdekat = i;
                    }
                }

                dataPersonaUser[simpanIndeks].listPersona[tempJumlah].nama =
dataPersona[terdekat].nama;
                dataPersonaUser[simpanIndeks].listPersona[tempJumlah].level =
dataPersona[terdekat].level;
                dataPersonaUser[simpanIndeks].listPersona[tempJumlah].arcana =
dataPersona[terdekat].arcana;
                    for(int i = 0; i <= dataPersona[terdekat].panjangSkill - 1;
i++){

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].skill[i] =
dataPersona[terdekat].skill[i];
                    }

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].panjangSkill =
dataPersona[terdekat].panjangSkill;
                    dataPersonaUser[simpanIndeks].jumlahPersona++;
                    tempJumlah++;

                    int maxIdx = max(persona1, persona2);
                    int minIdx = min(persona1, persona2);

                    for(int i = maxIdx; i < tempJumlah - 1 ; i++){
                        dataPersonaUser[simpanIndeks].listPersona[i] =
dataPersonaUser[simpanIndeks].listPersona[i + 1];
                        dataPersonaUser[simpanIndeks].jumlahPersona--;
                        tempJumlah--;
                    }
```

```
                    for(int i = minIdx; i < tempJumlah - 1 ; i++){
                        dataPersonaUser[simpanIndeks].listPersona[i] =
dataPersonaUser[simpanIndeks].listPersona[i + 1];
                        dataPersonaUser[simpanIndeks].jumlahPersona--;
                        tempJumlah--;
                    }

                }else {
                    cout << "nomor yang dipilih tidak ada" << endl;
                }

                break;
                case 4:
                if(dataPersonaUser[simpanIndeks].jumlahPersona == 0){
                    cout << "kamu belum punya persona apa apa" << endl;
                    break;
                }
                for(int i = 0; i <= tempJumlah - 1; i++){
                    cout << i + 1 << ". " <<
dataPersonaUser[simpanIndeks].listPersona[i].nama << endl;
                }
                cout << "pilih nomor persona : "; cin >> index;
                index -= 1;
                if (index >= 0 && index <= tempJumlah - 1) {
                    int tempSkill =
dataPersonaUser[simpanIndeks].listPersona[index].panjangSkill;
                        for(int j = 0; j < tempSkill ; j ++){
                            cout << j + 1 << ". " <<
dataPersonaUser[simpanIndeks].listPersona[index].skill[j] << endl;
                        }

                    cout << "pilih skill yang ingin diubah "; cin >> indexSkill;

                    if(indexSkill - 1 >= 0 && indexSkill <= tempSkill){
                        int indexSkillItem;
                        for(int i = 0; i < 5; i++ ){
                            cout << i + 1 << ". "<< skillItem[i] << " " << endl;;
                        }
                        cout << "pilih nomor skill baru : "; cin >> indexSkillItem;

                        if(indexSkillItem - 1 >= 0 && indexSkillItem <= 5){
                                for(int j = 0; j <= tempSkill - 1; j ++){

if(dataPersonaUser[simpanIndeks].listPersona[index].skill[j] ==
skillItem[indexSkillItem - 1]){
                                        jumlahDataSama = true;
                                    }
                                }
                                if(jumlahDataSama){
                                    cout << "sudah punya skill yang sama, batalkan
pembaruan" << endl;
                                }else{
```

```
dataPersonaUser[simpanIndeks].listPersona[index].skill[indexSkill -1] =
skillItem[indexSkillItem - 1];
                                    cout << "pembaruan berhasil" << endl;
                                }
                        }else {
                    cout << "nomor skill item tidak ada" << endl;
                }
                    }else {
                    cout << "nomor skill Persona tidak ada" << endl;
                }
                }else {
                    cout << "nomor Persona tidak ada" << endl;
                }
                break;
                case 5:
                if(dataPersonaUser[simpanIndeks].jumlahPersona == 0){
                    cout << "kamu belum punya persona apa apa" << endl;
                    break;
                }else{
                    for(int i = 0; i <= dataPersonaUser[simpanIndeks].jumlahPersona
- 1; i++){

                        cout << "\n ======" << i + 1 << ". "<<
dataPersonaUser[simpanIndeks].listPersona[i].nama << "======" << endl;
                        cout << "level    : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].level << endl;
                        cout << "arcana   : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].arcana << endl;

                        cout << "skill    : ";
                        for (int j = 0; j
<dataPersonaUser[simpanIndeks].listPersona[i].panjangSkill - 1; j++){
                            cout <<
dataPersonaUser[simpanIndeks].listPersona[i].skill[j] << ", ";
                        }
                    }
                }
                cout << "pilih persona yang ingin dihapus" ; cin >> index;
                if (index - 1 >= 0 && index - 1 <= tempJumlah - 1 ){
                    for(int i = index - 1; i < tempJumlah - 1; i++){
                        dataPersonaUser[simpanIndeks].listPersona[i] =
dataPersonaUser[simpanIndeks].listPersona[i + 1];
                    }
                }else{
                    cout << "nomor persona tidak ada" << endl;
                }
                break;
                case 0:
                cout << "log out" << endl;
                status = 0;
                break;

                default:
```

```
                cout << "Pilihan tidak valid." << endl;
                break;
            }
        }while(pilihan != 0);
```

## B. Fitur admin

pada kode ini terdapat semua fitur admin, mulai dari lihat data utama persona, hingga hapus persona dari data utama

```
if( status == 1){

            percobaan = 0;
            do {
              cout << "\n = menu velvet room admin =" << endl;
              cout << "1. tampilkan data persona" << endl;
              cout << "2. Tambah persona " << endl;
              cout << "3. update persona" << endl;
              cout << "4. Hapus persona" << endl;
              cout << "0. Keluar" << endl;
              cout << "Pilihan: ";
              cin >> pilihan;
              switch (pilihan) {
                case 1:
                if (totalPersona == 0){
                   cout << "tidak ada persona yang bisa dibaca" << endl;
                }else{
                for(int i = 0; i < totalPersona ; i++){
                    cout << "\n ======" << dataPersona[i].nama << "======" << endl;
                    cout << "level    : "  << dataPersona[i].level << endl;
                    cout << "arcana   : "  << dataPersona[i].arcana << endl;
                    cout << "skill    : ";
                    for (int j = 0; j < dataPersona[i].panjangSkill ; j++){
                        cout << dataPersona[i].skill[j] << ", ";
                    }
                    cout << endl;
                    cout << "harga   : "  << dataPersona[i].harga << endl;
                }
            }

                break;

                case 2:

                if (totalPersona < 20){
                    cout << "masukkan nama persona : ";
                    cin.ignore();
                    getline(cin, dataPersona[totalPersona].nama);

                    cout << "masukkan level persona : ";
```

```
                    cin.ignore();
                    cin >> dataPersona[totalPersona].level;

                    cout << "masukkan arcana persona : ";
                    cin.ignore();
                    cin >> dataPersona[totalPersona].arcana;

                    cout << "masukkan skill persona : " << endl;
                    cout << "ketik stop untuk berhenti : " << endl;
                    cin.ignore(1000, '\n');
                    int i = 0;
                    while(i <= 7){
                        string inputSkill;
                        cout << "skill " << i + 1 << ": "; getline(cin,inputSkill);

                        if (inputSkill == "stop"){
                        break;
                    }

                        dataPersona[totalPersona].skill[i] = inputSkill;
                        dataPersona[totalPersona].panjangSkill++;
                        i++;
                    }

                    cout << "masukkan harga persona : ";
                    cin.ignore();
                    cin >> dataPersona[totalPersona].harga;
                    totalPersona++;
                    cout << "data berhasil dimasukkan" << endl;
                }else{
                    cout << "kapasitas data penuh" << endl;
                }
                break;

                //update
                case 3:

                if (totalPersona == 0){
                   cout << "tidak ada persona yang bisa diubah" << endl;
                }else{
                    for(int i = 0; i < totalPersona ; i++){
                        cout << "persona ke " << i + 1 << " : " <<
dataPersona[i].nama << endl;
                    }
                    cout << "masukkan nomor persona yang ingin diubah" << endl ;
                    cin >> index;

                    if(index >= 1 && index <= totalPersona){
                        do {
                            cout << "1. ubah nama persona" << endl;
                            cout << "2. ubah level persona " << endl;
                            cout << "3. ubah arcana persona" << endl;
                            cout << "4. ubah skill persona" << endl;
```

```
                            cout << "5. ubah  harga persona" << endl;
                            cout << "0. batal update" << endl;
                            cout << "Pilihan: ";
                            cin >> pilihan;
                            switch (pilihan)
                            {
                            case 1:
                            cout << "masukkan nama baru : ";
                            cin.ignore();
                            cin >> dataPersona[index - 1].nama;
                            cout << "berhasil diubah" << endl;
                            break;
                            case 2:
                            cout << "masukkan level baru : ";
                            cin.ignore();
                            cin >> dataPersona[index - 1].level;
                            cout << "berhasil diubah" << endl;
                            break;
                            case 3:
                            cout << "masukkan arcana baru : ";
                            cin.ignore();
                            cin >> dataPersona[index - 1].arcana;
                            cout << "berhasil diubah" << endl;
                            break;
                            case 4:
                            for (int j = 0; j < dataPersona[index - 1].panjangSkill
; j++){
                                cout << "skill ke " << j + 1 << " : " <<
dataPersona[index - 1].skill[j] << endl;
                            }
                            cin.ignore();
                            cout << "masukkan nomor skill yang ingin diubah :"; cin
>> indexSkill;
                            indexSkill -= 1;
                            if(indexSkill >= 0 && indexSkill <= dataPersona[index -
1].panjangSkill - 1){
                                cin.ignore();
                                getline(cin,dataPersona[index -
1].skill[indexSkill]);
                                cout << "skill berhasil diubah" << endl;
                            }
                            break;
                            case 5:
                            cout << "masukkan harga baru : ";
                            cin.ignore();
                            cin >> dataPersona[index - 1].harga;
                            cout << "berhasil diubah" << endl;
                            break;

                            case 0:
                            cout << "batal update" << endl;
                            break;
```

```
                            default:
                            cout << "nomor tidak valid";
                            break;
                            }
                        } while(pilihan != 0);
                    }else {
                        cout << "tidak ada persona di nomor itu" << endl;
                    }
                }

                break;
                case 4:
                if (totalPersona == 0){
                   cout << "tidak ada persona yang bisa dihapus" << endl;
                }else{
                    for(int i = 0; i < totalPersona ; i++){
                        cout << "persona ke " << i + 1<< " : " <<
dataPersona[i].nama << endl;
                    }
                    cout << "masukkan nomor persona yang ingin dihapus" << endl ;
                    cin >> index;

                    if(index >= 1 && index <= totalPersona){
                        for(int i = index - 1 ; i < totalPersona - 1; i++){
                           dataPersona[i] = dataPersona[i + 1];
                        }
                        totalPersona--;
                        cout << "persona berhasil dihapus" << endl;
                    }else {
                        cout << "tidak ada persona di nomor itu" << endl;
                    }
                }
                break;
                case 0:
                cout << "log out...." << endl;
                status = 0;
                break;
                default:
                cout << "Pilihan tidak valid." << endl;
                }
            } while (pilihan != 0);
```

## 4. Hasil Output

```
 ======1. orpheus======
level     : 10
arcana    : fool
skill     : agi, dia, bash, tarunda,
harga    : 500

 ======2. jack o lantern======
level     : 15
arcana    : magician
skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
harga    : 750

 ======3. archangel======
level     : 20
arcana    : justice
skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
harga    : 1000
pilih nomor persona yang ingin dibeli :
```

```
 ======archangel======
level     : 20
arcana    : justice
skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
```

Gambar 4.1 opsi user(beli persona)

```
 ======1. archangel======
level     : 20
arcana    : justice
skill     : kouha, single shot, baisudi, sukukaja, holy arrow,

 ======2. orpheus======
level     : 10
arcana    : fool
skill     : agi, dia, bash,
masukkan persona pertama yang ingin digabung 1
masukkan persona kedua yang ingin digabung 2
```

```
 ======jack o lantern======
level     : 15
arcana    : magician
skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
```

Gambar 4.2 opsi user (fuse/gabung persona)

```
pilih skill yang ingin diubah 6
1. masukunda
2. dekaja
3. debilitate
4. charge
5. concentrate
pilih nomor skill baru : 5
pembaruan berhasil
```

```
 ======jack o lantern======
level     : 15
arcana    : magician
skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, concentrate,
```

gambar 4.3 opsi user (ubah skill persona)

```
 ======1. orpheus======
level     : 10
arcana    : fool
skill     : agi, dia, bash,

 ======2. jack o lantern======
level     : 15
arcana    : magician
skill     : bufu, pulinpa, mabufu, shock punch, freeze boost,
pilih persona yang ingin dihapus2
```

```
 ======orpheus======
level     : 10
arcana    : fool
skill     : agi, dia, bash, tarunda,
```

gambar 4.4 opsi user (hapus persona user)

```
Pilihan: 2
masukkan nama persona : mulyono
masukkan level persona : 22
masukkan arcana persona : aeon
masukkan skill persona :
ketik stop untuk berhenti :
skill 1: gorong gorong
skill 2: 11 ribu triliun
skill 3: stop
masukkan harga persona : 1900
data berhasil dimasukkan
```

```
 ======mulyono======
level     : 22
arcana    : aeon
skill     : gorong gorong, 11 ribu triliun,
harga    : 1900
```

gambar 4.5 opsi admin (tambah persona)

```
persona ke 1 : orpheus
persona ke 2 : jack o lantern
persona ke 3 : archangel
persona ke 4 : mulyono
masukkan nomor persona yang ingin diubah
4
1. ubah nama persona
2. ubah level persona
3. ubah arcana persona
4. ubah skill persona
5. ubah  harga persona
0. batal update
Pilihan: 3
masukkan arcana baru : psi
```

```
 ======mulyono======
level     : 22
arcana    : psi
skill     : gorong gorong, 11 ribu triliun,
harga    : 1900
```

gambar 4.6 opsi admin (edit persona)

```
Pilihan: 4
persona ke 1 : orpheus
persona ke 2 : jack o lantern
persona ke 3 : archangel
persona ke 4 : mulyono
masukkan nomor persona yang ingin dihapus
4
```

```
 ======orpheus======
level     : 10
arcana    : fool
skill     : agi, dia, bash, tarunda,
harga    : 500

 ======jack o lantern======
level     : 15
arcana    : magician
skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
harga    : 750

 ======archangel======
level     : 20
arcana    : justice
skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
harga    : 1000

= menu velvet room admin =
```

gambar 4.7 opsi admin(hapus persona)

## 5. Langkah-langkah GIT

### 5.1 GIT Init

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git init
Reinitialized existing Git repository in G:/praktikum/APL/.git/
```

gambar 5.1 inisiasi git

## 5.2 GIT Add

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git add /g/praktikum/APL/kelas/post-test/post-test-apl-2/2509106038-FEBRIANNO-
OZORA-ALINANTO-PT-2.cpp /g/praktikum/APL/kelas/post-test/post-test-apl-2/2509106
038-FEBRIANNO-OZORA-ALINANTO-PT-2.exe
```

gambar 5.2 menambahkan file baru ke git

## 5.3 GIT Commit

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git commit -m "submit posttest 2"
[main b1bf430] submit posttest 2
 2 files changed, 506 insertions(+)
 create mode 100644 kelas/post-test/post-test-apl-2/2509106038-FEBRIANNO-OZORA-A
LINANTO-PT-2.cpp
 create mode 100644 kelas/post-test/post-test-apl-2/2509106038-FEBRIANNO-OZORA-A
LINANTO-PT-2.exe
```

gambar 5.3 commit git local

## 5.4 GIT Remote

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git remote add origin https://github.com/ojola444/praktikum-apl.git
error: remote origin already exists.
```

gambar 5.4 git remote

## 5.5 GIT Push

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 85.86 KiB | 7.80 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ojola444/praktikum-apl.git
   cdd2c72..b1bf430  main -> main
branch 'main' set up to track 'origin/main'.
```

gambar 5.5 push commit local ke repository main